

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on source.efymag.com website.

the screen. The microcontroller stores the values of temperature, pressure and humidity every 30 minutes up to 35 hours. When this system is connected to PC via USB, all stored values of temperature, pressure and humidity are transferred to the GUI program. This program displays these values in the form of a graph on the PC screen. The 'time' is shown on the x-axis, and humidity, pressure and temperature are shown in y-axis as shown in Fig. 1.

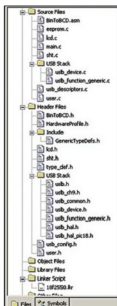


Fig. 3: MPLAB screenshot

Software

The software program for the microcontroller is written in 'C' language and compiled in MPLAB with C18 compiler. Add source file as shown in Fig. 3. The program is burnt in the MCU using a suitable PIC programmer.

Steps to run the Visual Studio program in PC (supports only 32-bit windows-operating system) are:

1. Download the USB drivers from Microchip website for PIC18F2550 microcontroller and install them in PC.
2. Connect the circuit to PC and confirm the connectivity in device manager.
3. Run executable file of Visual Studio application named 'Weather Logger.'
4. On the left-hand side, real-time values of temperature, pressure and humidity can be seen.

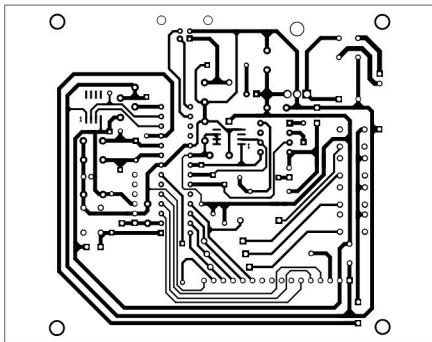


Fig. 4: An actual-size, single-side PCB for the weather logger

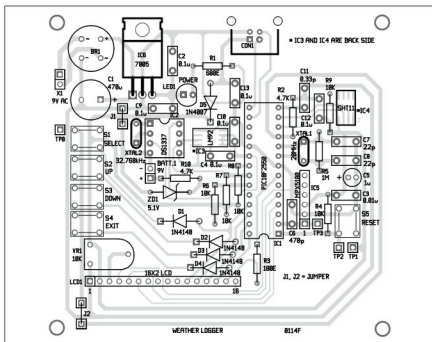


Fig. 5: Component layout for the PCB

Construction and testing

An actual-size, single-side PCB for the weather logger circuit is shown in Fig. 4 and its component layout in Fig. 5. Assemble the circuit on the recommended PCB to minimise assembly time and errors. Use IC bases for all the ICs. IC3 and IC4 are SMD ICs, so solder these ICs very care-

fully at the solder side of the PCB.

To test the circuit for proper functioning, verify 5V power supply at TP1 with respect to TP0. The reset to the system can be checked at TP2. ●

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